Day 1 Problem Set Solutions

Methods Camp | University of Texas at Austin

Casey Breen

Day 1 Problem Set

This document contains solutions to all exercises from the Day 1 problem set. To reinforce your understanding, please try your best to complete these exercises independently before referring to the solutions.

Exercise 1: Functions and Operations

1.1 Let $f(x) = 5x - 4$. Compute $f(2)$ and $f(0)$.

Solution: $f(2) = 5(2) - 4 = 6; \quad f(0) = -4$

1.2 Let $f(x, y) = 3x + y^2$. Compute $f(2, 3)$.

Solution: $f(2, 3) = 3(2) + 3^2 = 6 + 9 = 15$

1.3 Simplify: $\ln(50) - \ln(10)$. (Hint: use a log rule.)

Solution: $\ln(50) - \ln(10) = \ln(50/10) = \ln(5) \approx 1.609$

1.4 Solve for x: $\ln(x) = 2$. (Hint: rewrite in exponential form.)

Solution: $\ln(x) = 2 \Rightarrow x = e^2 \approx 7.389$

1.5 A class survey of 6 students records number of siblings: $\{1, 0, 2, 3, 1, 2\}$.

a. Write this as a summation expression and compute $\sum_{i=1}^{6} x_i$.

Solution: $\sum_{i=1}^{6} x_i = 1 + 0 + 2 + 3 + 1 + 2 = 9$

b. Compute the mean $\bar{x}$.

Solution: $\bar{x} = 9/6 = 1.5$

Exercise 2: Ratios, Proportions, Rates & Percentages

A town has 5,400 women and 4,600 men.

2.1 Compute the sex ratio (males per 100 females).

Solution: Sex ratio $= \frac{4{,}600}{5{,}400} \times 100 \approx 85.2$ males per 100 females.

2.2 Compute the proportion of the town that is female.

Solution: Proportion female $= \frac{5{,}400}{10{,}000} = 0.54$ (54%).

2.3 Last year the town recorded 27 burglaries. Compute the burglary rate per 100,000 residents. (Total population = 10,000.)

Solution: Burglary rate $= \frac{27}{10{,}000} \times 100{,}000 = 270$ per 100,000 residents.

2.4 The town's unemployment rate fell from 8% to 6%.

a. What is the change in percentage points?

Solution: Percentage-point change: $6 - 8 = -2$ p.p.

b. What is the percent (relative) change?

Solution: Percent change: $\frac{6-8}{8} \times 100 = -25\%$.

2.5 A local news headline says: "Unemployment drops 2%!" Is this headline accurate? Why or why not?

Solution: The headline is imprecise: the rate fell by 2 *percentage points*, not 2 *percent*. The true relative decline is 25%.

Exercise 3: Logarithms and Income

A researcher studies how income relates to years of education (X), using a model of log income:

$$\ln(Y) = 9.00 + 0.08X$$

where Y is annual income in dollars.

3.1 For a respondent with $X = 12$ years of education, compute $\ln(Y)$.

Solution: $\ln(Y) = 9.00 + 0.08(12) = 9.00 + 0.96 = 9.96$

3.2 Convert $\ln(Y)$ into Y (dollars) by exponentiating. (Hint: use the inverse of ln.)

Solution: $Y = e^{9.96} \approx \$21{,}159$

3.3 Now consider a respondent with $X = 13$. Compute their $\ln(Y)$ and their Y.

Solution: $\ln(Y) = 9.00 + 0.08(13) = 10.04$; $\quad Y = e^{10.04} \approx \$22{,}926$

3.4 Exponentiate the coefficient on X: $e^{0.08} \approx 1.083$. Interpret this number in one sentence.

Solution: $e^{0.08} \approx 1.083$: each additional year of education is associated with income being multiplied by about 1.08 — roughly an 8% increase in income, holding other factors constant.